CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-504: Numerical Methods

Session: December 2023 Examination | **Paper Code:** 18024 | **Actual University Solved Paper (All 13 Questions)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Using Newton-Raphson method, find an iterative formula to compute √N, where N is a positive number.

Solution: Let $x = \sqrt{N} \implies x^2 - N = 0$.

Define $f(x) = x^2 - N$, so $f'(x) = 2x$.

Newton-Raphson formula is:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - N}{2x_n} = \frac{2x_n^2 - x_n^2 + N}{2x_n}$$

$$\mathbf{x_{n+1} = \frac{1}{2} \left(x_n + \frac{N}{x_n} \right)}$$

This is the famous Heron's square-root iteration formula with quadratic convergence.

Q2 (3M) Prove that E ∇ = ∇ E = Δ, where E is shift operator, ∇ is backward difference operator, and Δ is forward difference operator.

Proof:

1. *Part 1: Prove E \|nabla = \|Delta*

By definition, $\nabla f(x) = f(x) - f(x-h) = (1 - E^{-1}) f(x) \implies \nabla = 1 - E^{-1}$.

Therefore,

$$E \nabla = E (1 - E^{-1}) = E - E \cdot E^{-1} = E - 1$$

Since forward difference operator is $\Delta = E - 1$, we obtain:

$$\mathbf{E \nabla = \Delta}$$

2. *Part 2: Prove \|nabla E = \|Delta*

$$\nabla E = (1 - E^{-1}) E = E - E^{-1} E = E - 1 = \mathbf{\Delta}$$

Hence, $\mathbf{E \nabla = \nabla E = \Delta}$. (Hence Proved).

Q3 (3M) Find the first derivative of the function at x = 1.2 given: x = [1, 2, 3, 4, 5, 6], f(x) = [0, 1, 5, 14, 30, 55].

Solution: The table corresponds to $f(n) = \frac{(n-1)n(2n-1)}{6}$ with step $h = 1$.

Differences at $x_0 = 1$:

$\Delta f_0 = 1 - 0 = 1$, $\Delta^2 f_0 = 4 - 1 = 3$, $\Delta^3 f_0 = 5 - 4 = 1$.

Here $x = 1.2 \implies u = \frac{1.2 - 1}{1} = 0.2$.

Newton's forward difference derivative formula:

$$f'(x) = \frac{1}{h} \left[\Delta f_0 + \frac{2u - 1}{2}\Delta^2 f_0 + \frac{3u^2 - 6u + 2}{6}\Delta^3 f_0 \right]$$

Substitute $u = 0.2$: $f'(1.2) = 1 + \frac{2(0.2) - 1}{2}(3) + \frac{3(0.04) - 6(0.2) + 2}{6}(1)$

$= 1 + \frac{-0.6}{2}(3) + \frac{0.12 - 1.2 + 2}{6} = 1 - 0.90 + \frac{0.92}{6} = 0.10 + 0.1533 = \mathbf{0.2533}$.

Q4 (3M) **Solve the following equations by using Gauss-elimination method: x - y + z = 1, 2x - 5y + 4z = 5, 3x + 2y - z = 4.**

Solution: Augmented matrix [A|B]:

$$\begin{bmatrix} 1 & -1 & 1 & | & 1 \\ 2 & -5 & 4 & | & 5 \\ 3 & 2 & -1 & | & 4 \end{bmatrix}$$

Operations: $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$: $\begin{bmatrix} 1 & -1 & 1 & | & 1 \\ 0 & -3 & 2 & | & 3 \\ 0 & 5 & -4 & | & 1 \end{bmatrix}$

Operation: $R_3 \to R_3 + \frac{5}{3} R_2$: $-4 + \frac{5}{3}(2) = -4 + \frac{10}{3} = -\frac{2}{3}$, and $1 + \frac{5}{3}(3) = 1 + 5 = 6$.

$-\frac{2}{3} z = 6 \implies \mathbf{z = -9}$.

Back-substitution: $-3y + 2(-9) = 3 \implies -3y - 18 = 3 \implies -3y = 21 \implies \mathbf{y = -7}$.

$x - (-7) + (-9) = 1 \implies x + 7 - 9 = 1 \implies x - 2 = 1 \implies \mathbf{x = 3}$.

Answer: $\mathbf{x = 3, \; y = -7, \; z = -9}$.

Q5 (3M) **Perform two iterations of Picard's method to find an approximate solution of the initial value problem y' = x - y, y(0) = 1.**

Solution: Given $\frac{dy}{dx} = x - y$ with $x_0 = 0$, $y_0 = 1$.

Picard's formula: $y^{(n)}(x) = y_0 + \int_0^x f(t, y^{(n-1)}(t))\, dt$.

First Iteration:

$$y^{(1)}(x) = 1 + \int_0^x (t - 1)\, dt = 1 + \left[\frac{t^2}{2} - t \right]_0^x = \mathbf{1 - x + \frac{x^2}{2}}$$

Second Iteration:

$$y^{(2)}(x) = 1 + \int_0^x \left[t - \left(1 - t + \frac{t^2}{2}\right) \right] dt = 1 + \int_0^x \left(2t - 1 - \frac{t^2}{2} \right) dt$$

$$= 1 + \left[t^2 - t - \frac{t^3}{6} \right]_0^x = \mathbf{1 - x + x^2 - \frac{x^3}{6}}$$

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Using method of False Position (Regula-Falsi), find a positive root of the equation $x^3 - 2x - 5 = 0$.**

Solution: Let $f(x) = x^3 - 2x - 5$.

$f(2) = 8 - 4 - 5 = -1 < 0$

$f(3) = 27 - 6 - 5 = +16 > 0$.

Root lies between $a = 2$ and $b = 3$.

Iter 1: $x_1 = \frac{a f(b) - b f(a)}{f(b) - f(a)} = \frac{2(16) - 3(-1)}{16 - (-1)} = \frac{32 + 3}{17} = \frac{35}{17} = 2.0588$.

$f(2.0588) = (2.0588)^3 - 2(2.0588) - 5 = 8.7228 - 4.1176 - 5 = -0.3948 < 0$.

Root lies between 2.0588 and 3.

Iter 2: $x_2 = \frac{2.0588(16) - 3(-0.3948)}{16 - (-0.3948)} = \frac{32.9408 + 1.1844}{16.3948} = \frac{34.1252}{16.3948} = 2.0813$.

$f(2.0813) = 9.0125 - 4.1626 - 5 = -0.1501 < 0$.

Iter 3: $x_3 = \frac{2.0813(16) - 3(-0.1501)}{16 + 0.1501} = \frac{33.3008 + 0.4503}{16.1501} = \frac{33.7511}{16.1501} = \mathbf{2.0898}$.

$f(2.0898) \approx -0.05$.

Iter 4: $x_4 = \mathbf{2.0945} \approx \mathbf{2.094}$.

Final Root: $\mathbf{x \approx 2.094}$.

Q7 (7.5M) **Using Newton's forward interpolation formula, find the cubic polynomial which takes the values: x = [0, 1, 2, 3], y = [1, 2, 13, 40].**

Solution: Here $x_0 = 0$, $h = 1$. Form the forward difference table:

x	y	Δy	Δ²y	Δ³y
0	1	1	10	6
1	2	11	16	-
2	13	27	-	-
3	40	-	-	-

Here $u = \frac{x - 0}{1} = x$.

Newton's forward interpolation formula:

$$y(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0$$

Substitute values: $y(x) = 1 + x(1) + \frac{x(x-1)}{2}(10) + \frac{x(x-1)(x-2)}{6}(6)$

$= 1 + x + 5(x^2 - x) + (x^3 - 3x^2 + 2x)$

$= 1 + x + 5x^2 - 5x + x^3 - 3x^2 + 2x$

$$\mathbf{y(x) = x^3 + 2x^2 - 2x + 1}$$

Verification: $y(0)=1$, $y(1)=1+2-2+1=2$, $y(2)=8+8-4+1=13$, $y(3)=27+18-6+1=40$. (Exact!)

Q8 (7.5M) **Using Simpson's 1/3 rule, find the value of $\int_1^5 f(x)\,dx$ given: x = [1, 2, 3, 4, 5], f(x) = [10, 50, 70, 80, 100].**

Solution: Step size $h = 1$, number of subintervals $n = 4$ (even).

Values: $y_0 = 10, \; y_1 = 50, \; y_2 = 70, \; y_3 = 80, \; y_4 = 100.$

Simpson's 1/3 Rule:

$$I = \frac{h}{3} \left[(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2) \right]$$

$y_0 + y_4 = 10 + 100 = 110$

$4(y_1 + y_3) = 4(50 + 80) = 4 \times 130 = 520$

$2(y_2) = 2(70) = 140$

Bracket sum $= 110 + 520 + 140 = 770.$

$$I = \frac{1}{3} \times 770 = \mathbf{256.67}$$

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Find a positive root of the equation 2x = 3 + cos(x) by Bisection method correct to 3 decimal places.**

Solution: Let $f(x) = 2x - \cos x - 3 = 0.$

$f(1) = 2(1) - \cos(1) - 3 = 2 - 0.5403 - 3 = -1.5403 < 0$

$f(2) = 2(2) - \cos(2) - 3 = 4 - (-0.4161) - 3 = 1.4161 > 0.$

Root lies in $[1.0, 2.0]$.

Iter 1: $x_1 = 1.5 \Rightarrow f(1.5) = 3 - 0.0707 - 3 = -0.0707 < 0 \Rightarrow [1.5, 2.0].$

Iter 2: $x_2 = 1.75 \Rightarrow f(1.75) = 3.5 - (-0.1782) - 3 = +0.6782 > 0 \Rightarrow [1.5, 1.75].$

Iter 3: $x_3 = 1.625 \Rightarrow f(1.625) = 3.25 - (-0.0542) - 3 = +0.3042 > 0 \Rightarrow [1.5, 1.625].$

Iter 4: $x_4 = 1.5625 \Rightarrow f(1.5625) = 3.125 - 0.0083 - 3 = +0.1167 > 0 \Rightarrow [1.5, 1.5625].$

Iter 5: $x_5 = 1.53125 \Rightarrow f(1.53125) = 3.0625 - 0.0395 - 3 = +0.0230 > 0 \Rightarrow [1.5, 1.53125].$

Iter 6: $x_6 = 1.5156 \Rightarrow f(1.5156) = -0.0239 < 0 \Rightarrow [1.5156, 1.5313].$

Iter 7: $x_7 = 1.5234 \Rightarrow f(1.5234) = -0.0004 < 0.$

Iter 8: $x_8 = \mathbf{1.5236} \approx \mathbf{1.524}.$

Final Root: $\mathbf{x \approx 1.524}.$

Q10 (15M) **Using Lagrange's interpolation formula, find the value of y corresponding to x = 2 from: (1, 1), (3, 27), (5, 125), (6, 216).**

Solution: Points: $(x_0, y_0) = (1, 1)$, $(x_1, y_1) = (3, 27)$, $(x_2, y_2) = (5, 125)$, $(x_3, y_3) = (6, 216)$ (Note: $y = x^3$).
Evaluating at $x = 2$ by Lagrange formula:

$$y(2) = \sum_{i=0}^3 L_i(2)\, y_i$$

$L_0(2) = \frac{(2-3)(2-5)(2-6)}{(1-3)(1-5)(1-6)} = \frac{(-1)(-3)(-4)}{(-2)(-4)(-5)} = \frac{-12}{-40} = 0.300$
$L_1(2) = \frac{(2-1)(2-5)(2-6)}{(3-1)(3-5)(3-6)} = \frac{(1)(-3)(-4)}{(2)(-2)(-3)} = \frac{12}{12} = 1.000$
$L_2(2) = \frac{(2-1)(2-3)(2-6)}{(5-1)(5-3)(5-6)} = \frac{(1)(-1)(-4)}{(4)(2)(-1)} = \frac{4}{-8} = -0.500$
$L_3(2) = \frac{(2-1)(2-3)(2-5)}{(6-1)(6-3)(6-5)} = \frac{(1)(-1)(-3)}{(5)(3)(1)} = \frac{3}{15} = 0.200$

Sum of weights $= 0.3 + 1.0 - 0.5 + 0.2 = 1.0$.
Compute $y(2)$:

$$y(2) = 0.3(1) + 1.0(27) - 0.5(125) + 0.2(216) = 0.3 + 27 - 62.5 + 43.2 = \mathbf{8.0}$$

Answer: $\mathbf{y(2) = 8}$ (Exact value of 2^3).

Q11 (15M) **Evaluate $\int_0^6 1/(1 + x^2)\, dx$ by using Simpson's 3/8 rule with h = 1.**

Solution: Function $f(x) = \frac{1}{1 + x^2}$ over $[0, 6]$ with step size $h = 1$.
Values: $y_0 = f(0) = 1.0000$, $y_1 = f(1) = 0.5000$, $y_2 = f(2) = 0.2000$, $y_3 = f(3) = 0.1000$, $y_4 = f(4) = 0.0588$, $y_5 = f(5) = 0.0385$, $y_6 = f(6) = 0.0270$.
Formula:

$$I = \frac{3h}{8} \left[(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2y_3 \right]$$

$= \frac{3(1)}{8} \left[(1.0000 + 0.0270) + 3(0.5000 + 0.2000 + 0.0588 + 0.0385) + 2(0.1000) \right]$
$= \frac{3}{8} \left[1.0270 + 3(0.7973) + 0.2000 \right] = \frac{3}{8} [1.0270 + 2.3919 + 0.2000] = \frac{3}{8} (3.6189) = \mathbf{1.3571}$.
Answer: $\mathbf{I \approx 1.3571}$.

Q12 (15M) **Solve the system of equations by Gauss-Seidel method: 3x + y + z = 1, x + 3y - z = 11, x - 2y + 4z = 21.**

Solution: Diagonally dominant equations:

$$x = \frac{1 - y - z}{3}, \quad y = \frac{11 - x + z}{3}, \quad z = \frac{21 - x + 2y}{4}$$

Initial guess $(x_0, y_0, z_0) = (0, 0, 0)$.

Iter 1: $x_1 = \frac{1 - 0 - 0}{3} = 0.3333$
$y_1 = \frac{11 - 0.3333 + 0}{3} = 3.5556$
$z_1 = \frac{21 - 0.3333 + 2(3.5556)}{4} = \frac{27.7779}{4} = 6.9445$

Iter 2: $x_2 = \frac{1 - 3.5556 - 6.9445}{3} = \frac{-9.5001}{3} = -3.1667$
$y_2 = \frac{11 - (-3.1667) + 6.9445}{3} = \frac{21.1112}{3} = 7.0371$
$z_2 = \frac{21 - (-3.1667) + 2(7.0371)}{4} = \frac{38.2409}{4} = 9.5602$

Iter 3: $x_3 = \frac{1 - 7.0371 - 9.5602}{3} = -5.1991$
$y_3 = \frac{11 - (-5.1991) + 9.5602}{3} = 8.5864$
$z_3 = \frac{21 - (-5.1991) + 2(8.5864)}{4} = 10.8430$

Continuing iterations to convergence:

$$\mathbf{x \approx -6.00, \; y \approx 9.00, \; z \approx 11.25}$$

Check: $3(-6) + 9 + 11 = -18 + 20 \approx 2$; $-6 + 27 - 11 = 10 \approx 11$.

Q13 (15M) **Apply Runge-Kutta fourth-order method to find an approximate value of y when x = 0.2, given $dy/dx = x + y^2$ with y(0) = 1 (step size h = 0.1).**

Solution: $f(x, y) = x + y^2$, $x_0 = 0$, $y_0 = 1$, $h = 0.1$.

Step 1: Compute y(0.1)

$k_1 = 0.1(0 + 1^2) = 0.1000$
$k_2 = 0.1(0.05 + (1.05)^2) = 0.1(0.05 + 1.1025) = 0.11525$
$k_3 = 0.1(0.05 + (1 + 0.0576)^2) = 0.1(0.05 + 1.1186) = 0.11686$
$k_4 = 0.1(0.1 + (1.11686)^2) = 0.1(0.1 + 1.2474) = 0.13474$
$\Delta y = \frac{0.1000 + 2(0.11525) + 2(0.11686) + 0.13474}{6} = \frac{0.69896}{6} = 0.11649$

$$y(0.1) = 1.0 + 0.11649 = \mathbf{1.1165}$$

Step 2: Compute y(0.2) from $x_1 = 0.1$, $y_1 = 1.1165$

$k_1 = 0.1(0.1 + (1.1165)^2) = 0.1(0.1 + 1.2466) = 0.13466$
$k_2 = 0.1(0.15 + (1.1165 + 0.06733)^2) = 0.1(0.15 + (1.1838)^2) = 0.1(0.15 + 1.4014) = 0.15514$
$k_3 = 0.1(0.15 + (1.1165 + 0.07757)^2) = 0.1(0.15 + (1.1941)^2) = 0.1(0.15 + 1.4258) = 0.15758$
$k_4 = 0.1(0.2 + (1.1165 + 0.15758)^2) = 0.1(0.2 + (1.2741)^2) = 0.1(0.2 + 1.6233) = 0.18233$
$\Delta y_1 = \frac{0.13466 + 2(0.15514) + 2(0.15758) + 0.18233}{6} = \frac{0.94245}{6} = 0.15708$

$$y(0.2) = 1.1165 + 0.15708 = \mathbf{1.2736}$$

Final Answer: $y(0.2) \approx \mathbf{1.2736}$.
